# 8. Implementation Plan — TCC Event Clearance App

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Build in six phases, pilot at one small department event, then scale to intramurals. Timings assume one student developer working part-time.

## Before you write code

The whole app depends on the first two items.

- [ ] Present the idea to your instructor and the SBO adviser; get a yes that a digital card counts as valid clearance, and agree how disputes are handled when a phone fails.
- [ ] Confirm the head SBO officer will pilot it with you.
- [ ] Brief the school's data protection officer and agree the consent text and retention rule.
- [ ] Send two or three more sample cards from other events or departments.
- [ ] Ask whether the school can pay for a paid backend plan if the free tier is not enough.
- [ ] Test the core idea on real phones (see Phase 0): non-extractable key, offline QR sign and verify, PWA install on one Android and one iPhone.
- [ ] Choose a small pilot event, such as a CIT event.

## Phases and milestones

Estimates assume one student developer working part-time and total about 14 to 20 weeks. They are guesses to revise after Phase 0.

| Phase | Scope | Done when | Weeks |
|---|---|---|---|
| 0. Validation | Install gate, key generation, offline QR sign and verify, camera scan speed on real phones | Works offline on at least one Android and one iPhone | 1 to 2 |

| Phase | Scope | Done when | Weeks |
|---|---|---|---|
| 1. Accounts | Google sign-in, registration, live photo with face check, department approval, device binding, reset requests, consent | A test student registers, is approved and gets a QR | 2 to 3 |
| 2. Events and cards | Event builder (days, slots, venues, signers), signature pad, card template renderer, presets, publish with approval record | The CIT Intramurals 2026 card is rebuilt from a template | 3 |
| 3. Scanner and check-in | Event pack and roster download, offline scanner, scan queue, sync function, duplicate handling, manual check-in, backup export | A full offline run with about 100 test students passes | 3 to 4 |
| 4. Stamps and checking | Signed records, stamp rendering, Check card, optional receipt QR, image and PDF export, disputes | A forged card shows slots as missing | 2 to 3 |
| 5. Web dashboard | Attendance by department, year and section, edits with audit log, Excel and PDF export, printing | A real attendance sheet prints correctly | 2 to 3 |
| 6. AI artwork | Edge function, artwork picker, generation limits | An SBO creates a card background in under 5 minutes | 1 to 2 |

## Pilot and rollout

1. **Pilot 1.** One small department event, one scanner, the paper process running in parallel as a safety net. Measure scan time, failures, sync problems and student feedback.
2. **Pilot 2.** Two or three departments with a scanner each, after fixing what pilot 1 showed.

3. **Registration drive.** Registration days per department with that department's SBO and school wifi, with a QR poster linking to the install page. Aim to have most students approved before intramurals.
4. **Intramurals.** Each department SBO runs its own scanner phone, with a spare phone and a power bank, and the head officer assigns the second scanner for the largest department. Paper cards stay available for manual cases.
5. **Day-before checklist for every officer:** scanner app opened, roster downloaded, "ready for offline" green, signature drawn, venue pins checked, phone charged.
6. **After each event.** Sync every scanner, export attendance, resolve disputes, collect feedback.

## Risks and fallbacks

| Risk | Fallback |
|---|---|
| The school does not accept digital cards | Run it as an attendance tool first, and print signed cards from the dashboard |
| Registrar or data protection concerns | Keep data minimal, use no registrar data, show consent, and delete data at year end |
| Too few students registered by the event | Registration days, plus manual check-in for walk-ups |
| Scanner phone dies or is lost | Power banks, a spare phone per department, revoke the lost scanner's key |
| Sync fails | Export the unsynced scans as a file |
| Solo developer runs out of time | Cut to the MVP, do AI artwork last, and use preset templates |
| Confusion installing on iPhone | Picture guide and a short orientation by the SBO |
| Free-tier limits hit | Compress artwork, cache packs, check limits early and budget for a paid plan |
| Students relay a QR from home | Photo comparison at the scanner, optional location check, school policy |